PUBLIC WORKING DRAFT · BUILD 20260906A

In progress, openly reviewable

This is a research snapshot, not a peer-reviewed or final publication. The underlying v1 manuscript preserves dated drafting notes for auditability; this banner carries the current status.

- Active statistical release: **second_pass_v6_675**, 675 modeled events.
- Public atlas: 370 authored cases and 380 playback stops.
- The 675-event release has 673 reviewed primary-family assignments and 2 explicit family evidence holds; held cases are not rendered in the atlas.
- The canonical analysis retains nine factors, with different retained counts under declared alternatives. The eight public traits summarize anchored coding judgments; the six families are reviewed editorial groupings. Neither is an automatic relabeling of the factors.
- Cards can include later source-reviewed display corrections awaiting a successor statistical release. The downloadable frozen v6 data do not silently incorporate those corrections or clear authoring holds.
- The normalized mechanism vocabulary failed its single predeclared blinded reliability retest and remains internal.
- Statistical inclusion is not automatic atlas inclusion: the atlas remains a separately authored 370-case playthrough.

Snapshot generated 2026-09-06T21:00:38Z. Corrections and case-specific feedback are welcome through the atlas feedback control.

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The Shape of Taking

An event register, a measured trait space, and the limits of automatic taxonomy in the world history of land-taking and removal, c. 6500 BCE to the present

Craig Talbert

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Craig Talbert *(sole author; AI systems credited as instruments in the contribution statement — decision recorded 2026-08-21; affiliation to add if desired)*

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PUBLIC WORKING DRAFT v1.5 — research update 2026-08-28; publication wording corrected 2026-09-05. This version is openly available for review and correction, but it is not peer reviewed or final. Supersedes v0 by applying the seven-item correction order in `MONOGRAPH_DRAFT_v0_FACT_CHECK.md` (independent findings-only audit, 2026-08-21). v0 is preserved unchanged for review. Bracketed `[TODO]` markers flag remaining production or verification work; the author decisions below are resolved. Alternate title: "*Taking Their Land: A Measured Register of Settlement, Displacement, and Removal.*"

Version 1.5 records the completed finite post-v5 correction cycle, its three-stage Editorial Suite, and the validated promotion of immutable `second_pass_v6_675`. The 2026-09-05 wording correction distinguishes current status from historical checkpoints; it does not change that research release.

Current frozen release and local publication snapshot (2026-09-05). The active statistical release is the immutable `second_pass_v6_675`: 675 modeled events, of which 673 have reviewed primary-family assignments and two remain explicit family evidence holds. Its canonical preprocessing retains nine exploratory factors; the retained count and factor space remain sensitive to declared alternatives (§5.2). The eight-trait analytical profile summarizes hand-coded anchored judgments (AI-assisted as disclosed in §4), not the nine-factor solution. The six primary families are source-reviewed editorial groupings, not discovered clusters. These are three different products, not competing counts of the same thing.

The local atlas contains 370 authored cases and 380 playback stops at the `20260905a` checkpoint; `20260905b` is the coordinated publication-refresh target. Statistical promotion does not silently render unauthored cases, and a local build does not establish live deployment. No live verification was performed for this wording correction. The two current-state authorities are `CURRENT_WORK_STATE.md` and `analysis/second_pass/PIPELINE_STATE.json`; `analysis/second_pass/CONTENT_STATISTICAL_OPEN_LOOPS.md` records the substantive backlog.

Frozen data are not the post-freeze display layer. Later source, boundary, and public-card corrections through Batch 50 do not mutate v6. Reviewed display overrides may therefore prevent a displayed card from reproducing from frozen v6 alone. Accepted corrections still require reconciliation, any required coding, and a separately gated successor fit before statistical promotion. Authoring holds and unresolved successor events remain held; the two frozen family holds are not a count of all workflow holds. Robustness results below belong to their named releases, not to unintegrated corrections or held events.

Public working-draft authorization (2026-08-24). The author approved an HTML publication of this in-progress manuscript and a checksum-bound download of the then-active v4 release data. The public page carries a conspicuous status banner and links to the exact Markdown source. The 2026-08-27 refresh replaced that data package with promoted v5. Publication build `20260828z` first served v6 alongside an atlas of 313 authored cases and 323 playback stops; those are historical build counts, not the current local inventory. Copyrighted source captures, scratch files, and raw model/coder journals remain excluded from the public bundle.

Event-unit fairness requirement added 2026-08-23. Monograph v2 must make explicit that comparable events are not equal-duration time boxes. It must report a release-wide boundary audit and alternative-resolution sensitivity tests before claiming that one-event/one-row treatment is consistently applied across historical tempos and archive densities. The frozen prospective method and audit schema are in `analysis/second_pass/EVENT_UNIT_FAIRNESS_AND_RESOLUTION_PROTOCOL.md` and `analysis/second_pass/event_unit_fairness_audit_template.csv`; the proportionate execution and stop rule are in `analysis/second_pass/event_unit_fairness_v1/STREAMLINED_EXECUTION_PLAN.md`. The working subsection below records the argument and the completed release-level diagnostics. Coverage completeness is not claimed. Release activation is complete under the qualified decision; the mechanism/family and public-card decisions are closed for this release as described below.

An early 100-row corpus-only check opened no source passages and was disqualified rather than counted as independent historical verification. The replacement review opened evidence for all 333 preregistered event checks and 245 relationships; §2.2.1 records the completed process and its limits.

Historical event-unit candidate-model checkpoint (2026-08-27; corrected and promoted as v5). The first 583/679 assembly is preserved as a superseded diagnostic. Its builder failed to consume three source-resolved territorial-scope exclusions already documented elsewhere in the draft and review record: Tibetan residential-school separation, the Swiss Pro Juventute removal of Jenisch children, and the Svalbard evacuation. The corrected internal arithmetic is `583 − 3 = 580` for the parent reference and `580 − 61 + 157 = 676` for the successor candidate. Falklands/Malvinas remains included as the positive land-taking comparator. The candidate was first materialized without changing v4 and was then promoted as immutable v5 without altering the prior release. The corrected model battery reproduced nine factors under every one-event deletion and in twenty independent Horn batches; every nine-dimensional leave-one-event subspace angle remained below five degrees (maximum 4.808). Whole-program, date-proxy, weighting, imputation, documentation, and source-resolution alternatives remain material. The result rules out single-row domination; it does not approve every event boundary or settle the factor count. The later Editorial Suite and explicit `PROMOTE_WITH_QUALIFICATIONS` decision authorize the release while preserving those limits.

Historical mechanism/family checkpoint (2026-08-27; terminal for v5). The audit-corrected candidate-676 model battery found substantial association but remaining distinction between process tags and structural outcome families. Those estimates remain a historical diagnostic, not a public-label result: after one bounded source-first repair, the single permitted blinded retest failed its predeclared reliability gate. Section 6 reports both results and the stopping rule. Normalized mechanism tags remain internal; primary family and the evidence-aware trait profile remain the public analytical layers. The mechanism diagnostic itself grants no atlas or deployment authority.

Post-v5 correction checkpoint (2026-08-28; promoted as v6). A finite post-v5 cycle applied ten source-adjudicated authoring corrections and compared the resulting model. The candidate retires the broad Indo-Aryan migration wrapper to an inactive audit row, applies 117 adjudicated cell changes across nine retained events, corrects two chronology-misleading Assyrian identifiers through durable aliases, and contains 675 active events.

V5 and the candidate both retain nine factors. Their nine-dimensional spaces differ by a maximum principal angle of 2.765 degrees; forced eight moves 6.685 degrees and rotates the allocation of three named axes. None of the nine leave-one-corrected-event-out scenarios crossed the declared materiality thresholds. This supports a stable nine-factor subspace with forced-eight axis rotation, not a claim that F2, F7, or F8 vanished. Existing weighting, missingness, imputation, documentation, and coherent-boundary sensitivities remain. The candidate passed the final Editorial Suite and the staged immutable-release validator, and is now active as `second_pass_v6_675`. Immutable v5 remains preserved.

Companion site: *Taking their land: a chronological atlas of settlement and displacement*, https://latrian.dyndns.org/takingtheirland/

Change log v0 → v1

Publication-wording correction (2026-09-05). Current status now identifies frozen v6/675 and the local 370-case/380-stop atlas separately from historical v5/676 and older atlas builds. Dated results and corrections remain intact. The introduction and conclusion foreground documented recurrence within a selected-positive-case corpus, not universal human inevitability or propensity. The historical 407-event report and its template now carry a superseded notice linking the working monograph and data downloads. No research values, licensing terms, holds, or release authority changed.

1. Factor-count claims in the original v1 correction: eight factors at 407 and 490 events; **nine** in the then-current v3 583-event roster by a +0.019 margin, characterized in §5.2; the forced-eight 407→583 congruence range (0.653–0.958) replaces the transplanted 407→490 range.
2. "No natural species" demoted from finding to explicitly editorial metaphor; the defensible result is stated as "no tested automatic partition is robust enough to serve as the public taxonomy."
3. §4 reframed as an *agreement-tested* protocol: cross-model N/A concordance (0.50 over 96 cells) reported; missing raw validation inputs disclosed; unsupported attenuation claim replaced.
4. Bibliography claims reduced to what the saved generator reproduces: unique **URL records** (845 at the audited checkpoint; 854 on that regeneration: +7 audit-register sources from the Sobaipuri/Salinas admissions, +2 from the recode-file harvest added in the second review), with known defects enumerated; the unreproducible ~196-locus count retired.
5. Transcript-loss claim reversed: five surviving session files are inventoried; the bounded process-fact recovery pass is complete.
6. At the original v1 checkpoint, current-state text was updated to atlas build `20260821a` (222 cases, 285 phase stops). The current public state is given above rather than silently rewriting that dated change-log fact.
7. Cohen 2013 re-placed (Hellenistic east, not Roman Italy); Seymour 2012 subtitle corrected (*Migrations*).

Second-review corrections (2026-08-21, same day): gap-scan arithmetic fixed (33 confirmed = 30 initial + 3 late; 12, not 15, remain unverified); raw validation journals recovered and copied into `analysis/validation/raw/` with both comparisons recomputed exactly from the in-project copies, resolving former limitation 3; census generator patched to harvest recode files (854 unique URL records); "well-documented modern cases accumulate" corrected to composition/resolution language (the 490→583 delta was 95 ancient Roman cases); "orthogonal" softened to "largely distinct axis" (oblique factors correlate); "published scholarship" broadened to "published sources."

Post-review completions (2026-08-21): the three remaining work items outside the author's own questions are done. (a) *Transcript recovery*: the bounded pass over the five Codex sessions is complete — six newly recorded

process facts, twelve decision-provenance confirmations with locators, and three negative findings, in `analysis/second_pass/PROCESS_FACT_RECOVERY.md`; §4.5 updated. (b) *Citation resolution*: the actual 40-entry v1 list is fully verified — 36 correct without change, 4 corrected, 0 unverifiable (`monograph/CITATION_RESOLUTION.md`). The original two-agent split covered 39; a closure audit caught and independently checked the omitted Ziems 2024 entry. (c) *Work-level bibliography*: `build_work_level_bibliography.py` maps the 854 URL records to 852 conservative pre-review groups. All 14 near-title pairs are adjudicated (12 different works, 1 same work, 1 part-of), producing 851 reviewed work groups (544 titled works, 155 primary loci, 122 DOI-keyed, 30 modern slot records). Typed publication metadata remains a research pass, not a script. The review also corrected a Habsburg source attribution from Alexander Szakolczai to Bogdan G. Popescu; see `monograph/BIBLIOGRAPHY_REVIEW.md`.

Author decisions — all resolved 2026-08-21

1. **Authorship:** Craig Talbert, sole author; AI systems credited as instruments in the contribution statement, not as co-authors.
2. **Citation style:** Chicago author–date, confirmed.
3. **Cunningham's Law:** named in §8 (the framing is the author's own, per the recovered decision stream).
4. **Bibliography form:** the citation-resolved 40-entry anchor list stays in the monograph; one later source-audited case citation (Malhi et al. 2008) is added rather than left dangling. The reviewed 851-work inventory publishes separately as a versioned dataset with a methods preface.

Closed earlier: the six working family names were confirmed in the recovered decision stream; the 40-entry citation pass and four corrections are verified (the defensible Belich title-page form is retained); all 14 bibliography relationships are adjudicated; and the exact unlinked report URL and historical deployment checksum are recorded in `analysis/second_pass/PROCESS_FACT_RECOVERY.md`.

Remaining production work (no author decision blocking): typed bibliographic metadata for the 851-work dataset and an archival deposit before formal circulation. Version control is established; current local publication status is stated in the opening snapshot, separately from live verification.

Abstract

Comparative discussion of settler colonialism, ethnic cleansing, and forced displacement relies on categorical vocabularies — "settler colonialism," "internal colonization," "ethnic cleansing" — whose boundaries were set by theory and case tradition rather than by systematic comparison. This monograph describes the construction of an alternative: a worldwide register of land-taking and removal events (675 modeled events in frozen statistical release v6; 370 authored cases in the companion atlas's local checkpoint), each coded on sixty ordinal dimensions drawn without merging from five disciplinary traditions, with every event researched against published sources. Coding was performed by large-language-model research agents under an **agreement-tested protocol with known cross-model disagreement**: a two-coder pilot reached 93% exact agreement over 345 comparisons; a candidate coder model was excluded after failing a validation gate (67% exact; 22% concordance on the load-bearing distinction between "unknown" and "structurally inapplicable"); cross-model agreement on a blind 30-event overlap was 76% exact and 95% within one anchor step, but only 50% concordant on the unknown-versus-inapplicable boundary. The sixty dimensions contain a **persistent multidimensional trait structure whose exact factor count is boundary-sensitive**: parallel analysis retains eight factors at 407 and

490 events, retained nine in the earlier v3 583-event release by a small margin (+0.019), and returned to eight in v4's 583-event release with the ninth Horn margin just below zero (−0.00789). The historical v5 676-event release retained nine in twenty repeated Horn calibrations and under every single-event deletion, but returned to eight under several historically plausible grouping and date-proxy alternatives. Frozen v6 retains nine, with its nine-dimensional space 2.765 degrees from v5; the exhaustive 676-event deletion battery is a v5 result, not a new v6 test. The boundary pattern is therefore not a clean additional public trait. Clustering on the trait space yields no tested automatic partition robust enough to serve as public categories: partitions are shallow, near-tied across hundreds of restarts, and sensitive to archive density — replacing one unusually granular source archive (the early Neo-Assyrian royal inscriptions) with resolution-matched medians moves cluster boundaries substantially while the factor structure barely shifts. We describe the editorial consequences implemented in the atlas — a manually reviewed six-family filing system, per-case eight-trait evidence-aware profiles that summarize coded judgments rather than latent factors, versioned immutable releases, case retirements with preserved audit rows, and a public correction channel. A normalized mechanism vocabulary failed its single permitted blinded reliability retest and therefore remains internal. We argue that the resulting measured-traits-plus-reviewed-families architecture is a replicable template for large-scale comparative history built with AI assistance under human editorial control. This selected-positive-case corpus documents recurrence and supports comparison; it cannot establish universal human inevitability or propensity to take land or remove people.

Keywords: settler colonialism; forced displacement; ethnic cleansing; comparative history; event data; exploratory factor analysis; cluster stability; large language models; AI-assisted coding; digital history.

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1. Introduction

Some historical processes move people onto land; others move people off it; many do both at once. The scholarly vocabularies for these processes grew up separately — settler-colonial studies around the permanence of incoming populations (Wolfe 2006; Veracini 2010), genocide and forced-migration studies around destruction and flight (Lemkin 1944; Naimark 2001; Mann 2005), political economy around land and labor regimes (Nieboer 1900; Domar 1970; Acemoglu, Johnson, and Robinson 2001), historical demography and archaeogenetics around measurable population turnover (Haak et al. 2015; Reich 2018), and the study of state formation around sovereignty, legibility, and borders (Scott 1998; Tilly 1990; Maier 2016). Each vocabulary illuminates the cases it was built on and strains on its neighbors'. The result, familiar to any comparative reader, is a taxonomy problem: categories that behave less like discoveries about the world than like the disciplinary histories of the people who coined them.

The central historical claim is recurrence: across widely separated times and places, documented land-taking and forced removal recur in recognizable combinations of power, movement, exclusion, settlement, and institutional change. Comparing those combinations reveals recurring processes and consequential differences. This selected-positive-case register has no representative denominator of all human encounters or opportunities for taking; it cannot establish a universal human propensity, inevitability, or the rarity of peaceful alternatives.

This monograph accompanies *Taking their land*, a public chronological atlas of settlement and displacement from roughly 6500 BCE to the present. The atlas began, as such projects usually do, with a hand-built category scheme — twelve mechanism families assigned case by case. It now rests on something different, and the difference is the subject of this paper. Rather than adopting any one discipline's framework, or negotiating a compromise between them, we treated the taxonomy itself as an empirical question: **if every event is measured on every tradition's dimensions at once, what structure actually emerges?**

Three features of the resulting study are worth stating at the outset, because each is a deliberate methodological commitment:

First, the unit of analysis is the bounded event, not the national story. A "unique historical event" is one linked process with substantially continuous actors, mechanism, geography, and chronology; processes are split when a regime changes, a mechanism changes materially, or a long interruption makes continuity misleading (§2). Much of the project's labor — and several of its most consequential editorial decisions, including the retirement of already-published atlas cases — consisted of enforcing this rule against the gravitational pull of familiar multi-century umbrellas.

Second, measurement was performed at a scale, and by a means, that requires disclosure and validation rather than apology. Every one of the 675 events in frozen v6 carries sixty anchored codes, including explicit unknown and structurally inapplicable states; every code was assigned by an AI research agent working from a written anchor protocol and per-event source research, inside an agreement-testing regime — pilot double-coding, a model-selection gate that rejected one candidate model on measured grounds, and cross-model agreement estimation — described with its known gaps in §4. We treat "AI-assisted comparative coding under human editorial control" as a methodological contribution in its own right, with honest agreement numbers and honest holes attached.

Third, the analysis was allowed to fail, and the write-up is required to keep the failures. The trait structure persisted as the corpus grew; the exact factor count did not (eight at 407 and 490, nine in v3's 583-event roster, eight in v4's different 583-event roster, and nine in both the historical 676-event v5 and current frozen 675-event v6 releases under their canonical preprocessing — §5.2), and no tested clustering of the trait space survived its own diagnostics as a public taxonomy (§6). We regard the negative results as equally important, particularly for the growing genre of computational history that reaches for cluster labels as if they were facts about the past rather than facts about a corpus, an encoding, and an algorithm (cf. Hennig 2015 on the method-dependence of "true" clusters). The process was iterative rather than a single pre-planned protocol; §5–§6 therefore distinguish explicitly between what was specified before analysis, what was found exploratorily, and what was decided editorially afterward.

The monograph is written in two registers. The main text is intended to be readable by a serious general reader of the atlas; technical material — estimation choices, stability statistics, sensitivity designs — is confined to marked subsections and tables, and the complete apparatus (data, scripts, manifests, hashes) is cited by file throughout (§10, Data availability).

1.1 Contributions

1. A worldwide, source-bound register of land-taking and removal events (675 modeled in frozen v6), with every row in the audit's declared country-and-territory sweep closed either by an event finding or a source-backed negative finding, and a source inventory of 854 unique URL records at the 2026-08-21 census checkpoint (§2.5).
2. A sixty-dimension measurement instrument assembled *without merging* from five disciplinary traditions, with anchored ordinal scales, era-coverage rules, and an explicit distinction between "unknown" and "structurally inapplicable" — the latter itself analytically load-bearing.
3. An **agreement-tested protocol** for AI-agent historical coding: pilot agreement 0.93 exact over 345 comparisons; a failed-model exclusion gate; cross-model agreement 0.76 exact / 0.95 within-1 on a blind 30-event overlap, with a disclosed weak point (0.50 N/A concordance). The raw inputs that were missing at the first audit were recovered and both comparisons now reproduce (§4.4).

4. A persistent multidimensional trait space with a documented **boundary-sensitive factor count** (eight at 407 and 490; nine in the v3 583-event roster by +0.019; eight in the v4 583-event roster; and nine in the historical v5 676-event and current frozen v6 675-event releases under their canonical preprocessing), a documented era confound on one axis, and a documented face-validity correction to another.
5. A demonstrated failure of automatic taxonomy *as tested*: shallow, near-tied partitions (367 distinct solutions in 500 restarts at 490 events) and a controlled archive-density experiment showing that one dense source archive relocates cluster boundaries in this corpus.
6. An editorial architecture that metabolizes (1)–(5) into a public artifact: reviewed families as filing (not discovery), evidence-aware trait profiles summarizing coded judgments (not latent factors), versioned immutable releases, case retirement with preserved audit rows, and an in-atlas correction channel.

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2. The register

2.1 From atlas audit to worldwide register

The register began as an audit of the atlas's own coverage. In August 2026 the then-current atlas contained 106 cases in 119 display phases. The audit posed five questions for every one of 196 sovereign states and 58 additional territories and cross-border homelands — who displaced, who was displaced, whether the atlas captured each process, what was missing, and what related processes fell outside the atlas's threshold — and closed all 254 geographic reviews, six of them with source-backed negative findings (Andorra, Liechtenstein, San Marino, Vatican City, American Samoa, Wallis and Futuna) rather than silence. The audit's deduplicated master register contained 364 events with 1,559 country-event incidences; its event-unit test found that only 36 of the 106 atlas cases survived unchanged, with 37 incomplete, 27 wrongly bundled, and 6 wrongly classified. Applying the audit produced a 222-case, 283-phase atlas build. (Register, methodology, and per-country findings: `audit/`, esp. `MASTER_EVENT_REGISTER.md`, `METHODOLOGY.md`, `EXECUTIVE_FINDINGS.md`.)

Two commitments from the audit govern everything downstream. **Modern states are geographic containers, not actors:** an "agent" is a historically named state, institution, army, or population stream, never a present-day population; counts never imply collective guilt. **Categories describe mechanisms, not moral severity:** the register's twelve mechanism families (settler colonization; conquest with settlement; expulsion/deportation/ethnic cleansing; demographic expansion; state resettlement/internal colonization; conflict displacement; genocide-linked clearance; enslavement/labor removal; development/conservation displacement; population exchange/partition; conquest without substantial settlement; return/decolonizing displacement) rank nothing.

2.2 The event-unit rule

The register's single most consequential instrument is the event-unit rule quoted above (§1). Its effect is easiest to see in what it dismantled. The audit split the atlas case "Anatolian Neolithic" into regionally distinct Neolithic expansion events; the second research pass (§2.4) split a single c. 900–612 BCE "Neo-Assyrian deportations" umbrella into eleven bounded successors, then added forty-four separately bounded earlier episodes (sixteen under Ashurnasirpal II, twenty-four under Shalmaneser III, four under Šamši-Adad V) from a seventy-five-record ruler-by-ruler ledger of the royal inscriptions (Oded 1979; project ledgers `EARLY_NEO_ASSYRIAN_SOURCE_REVIEW.md`, `_EVENT_UNITS.md`); and the third statistical release retired two published Roman wrapper cases (`med-roman-republican-colonies-italy`, `med-roman-po-valley-colonization`) in favor of

ninety-five bounded colonization events assembled from a 340+-disposition Greco-Roman source ledger anchored in the ancient narrative and antiquarian sources (Livy, Dionysius, Appian, Velleius; Salmon 1969). The individual classical passages and editions are resolved in that source ledger; the four author names are not claims of one-to-one entries in the selected modern reference list. The same rule, applied in the other direction, retired the atlas's broad "Southern Athabaskan migration" case: a well-evidenced four-century migration and ethnogenesis that never resolves into one demonstrated displacement event. Its bounded successors — the 1762 Sobaipuri-O'odham relocation, the 1667–1672 Las Humanas abandonment, and the 1676–1677 Quarai/Chililí-to-Tajique relocation — passed candidate review and are admitted, map-authored v4 cases with completed sixty-dimension coding. This is a deliberately public example of the rule outranking a published case (`analysis/second_pass/CURRENT_ATLAS_ATHABASKAN_REVIEW.md`, `SOUTHWEST_BOUNDED_SUCCESSOR_ADMISSION.md`).

2.2.1 Event-unit fairness across historical tempos

Equal statistical weight is meaningful only if the rows have been bounded by equivalent rules. It does not follow that they should occupy equal amounts of calendar time. A population transition visible only as a broad archaeological horizon around 6000 BCE, a campaign reported season by season in 338 BCE, and a removal documented day by day in 2014 differ in historical tempo and in the resolving power of their sources. Dividing all three into fixed year windows would reward modern documentation and manufacture comparability rather than achieve it.

The project's unit is therefore causal and territorial, not merely chronometric. Chronology is one test alongside continuity of actors or program, affected population and land, mechanism, causal sequence, geography, and outcome. A long interval prompts review but does not force a split; actions close in time may still be different events. A proposed child must have its own bounded coercive act or land transfer, affected population or tract, evidence for an independent analytical profile, and an explicit supersession rule. A removal, successor settlement, and later return may remain phases of one event when they implement one linked transformation; several map stops never become several statistical rows merely because the animation needs them.

The source-qualified boundary review supplies several useful comparisons. The Habsburg Military Frontier remains one 1522–1881 event because a named land-for-service military institution links its changing districts and populations. By contrast, the much shorter Balkan Wars, Croatian War, and Bosnian War wrappers do not become single events merely because each occupies one war interval: their source bodies identify independently organized programs, target populations, directions, and territorial purposes. The 1991–2008 South Ossetia wrapper likewise splits across the 1992 settlement and the newly constituted 2008 war, while the four-day Osh outbreak remains one event because Jalal-Abad's violence followed Osh within a contiguous regional mobilization and displacement episode. These paired decisions show why neither long duration nor short duration determines the unit.

A later Melos–Veii–Circeii authoring review makes the weighting consequence especially concrete. Veii remains one 396–387 BCE row because conquest, captive sale, household land assignment, selective citizenship, and tribal reorganization form one territorial program, even though its phases and cohorts must remain visible. Melos remains two oppositely directed events in 416 and 405 BCE because authority, affected cohort, direction, coercive act, and land outcome reverse; the broad parent remains inactive. Replacing those two Melos children with the parent rotates the forced-eight subspace by 10.81 degrees but the corrected nine-factor subspace by 2.92 degrees, while omitting the restoration alone moves them by only about 1.33 and 1.25 degrees. The source-first split therefore survives, but it is not presented as weight-neutral. Circeii supplies a different boundary: one named 393 BCE colony passes the declared durable-taking/use rule without proof of resident expulsion, while an unnamed 395 authorization and a source-conflicted earlier removal remain outside the event. These are

applications of one causal-territorial rule, not equal-duration boxes or archive-density quotas (ATLAS_AUTHORING_BATCH_33_ADJUDICATION.md).

Thebes supplies a second material-sensitivity example for the final v2 fairness account. Alexander's 335 BCE destruction and Cassander's 316 BCE restoration remain separate because authority, moving cohort, direction, and land outcome reverse after nineteen years; the primary sources independently bound both actions, and the inactive parent cannot coexist with both successors. Yet replacing the successors with that parent rotates the forced- eight subspace by about 5.60 degrees, while leaving the program out rotates it about 5.02 degrees. The corresponding corrected nine-factor movements are about 2.46 degrees, and nine factors remain retained. The split is therefore historically defensible but not weight-neutral. Reporting both results is the fairness obligation: the statistical consequence cannot be hidden, and it cannot be allowed to collapse two opposed historical processes merely to make the model move less (ATLAS_AUTHORING_BATCH_37_ADJUDICATION.md).

The same test works in the other direction. Darfur can remain a long phased event only after its scope is narrowed to the documented government/Janjaweed/Border Guards/RSF demographic-clearance program against non-Arab communities, including land occupation and obstruction of return. Unrelated SAF–RSF combat, rebel abuses, inter-Arab violence, and general Darfur displacement are outside that row. Eastern Congo and Somalia, despite equally persistent regional crises, require splits because their actors, mechanisms, affected populations, causal systems, and territorial programs vary independently. Liberia splits across the 1997 political settlement, return episode, and changed actor roles. The Highland Clearances illustrates the fail-closed alternative: evidence can disprove one national score profile without yet supplying a complete admitted successor set, so the wrapper stays on boundary hold rather than being counted or prematurely retired as a parent. These dispositions show the rule in operation but do not themselves authorize a release. Darfur's narrowed scope still required a compatibility check; the completed candidate counts and sensitivity results are reported below.

Project continuity, rather than calendar span, produces the same distinction elsewhere. The Three Gorges resettlement remains one event despite long, staged, multijurisdictional implementation because one state project, approved plan, inundation zone, funding system, and irreversible land outcome link its movements. Canada's 1953 and 1955 High Arctic relocations likewise remain one kin-linked federal project rather than two events defined by calendar year. By contrast, the much shorter 1978–1990 Nicaragua wrapper splits at the July 1979 regime change: most of the earlier exile cohort returned, new cohorts departed, authorities and directions changed, and the later Contra cycle included independently profileable Miskito relocation. The Australia-wide Stolen Generations category passes the atlas's territorial-scope threshold because the national inquiry documents severance from Country and impairment of land and native-title connections, but it still fails as one statistical unit because distinct jurisdictions used different laws, authorities, procedures, placements, and timelines. Scope qualification, event coherence, calendar duration, and geographic size are therefore separate judgments. The same territorial threshold excludes both Tibetan residential-school separation and the Swiss Pro Juventute removal of Jenisch children from active counting: each is a coherent or recognizable assimilation history, but the opened evidence establishes family separation and institutional placement rather than a territorial land outcome. Their exclusion and Australia's scope qualification are different evidentiary results under one rule, not different rules for different countries.

Wartime cases separate event disaggregation from scope admission. Saint-Pierre's British removals in 1778 and 1794 are distinct events because restoration and actual resident return reversed the first outcome before a new war produced the second. The Isle of Man's 1914–1919 and 1940–1945 internment systems likewise split after complete release, restoration of civilian premises, and a new activation involving materially different cohorts. But splitting the Channel Islands wrapper does not make its every component admissible: the substantially voluntary 1940 protective evacuation is contextual history, whereas Nazi deportation and internment must pass the

territorial-removal and overlap gates independently. Svalbard supplies the harder comparator. Its evacuation was compulsory, but it remains outside scope because it was a temporary wartime denial measure without a successor population or durable land transfer, followed by postwar re-establishment. Shared war, shared place, short duration, and even coercion are therefore not stand-alone boundary or scope rules.

Prehistoric and ancient cases expose the converse risk: turning an evidence limit into an unannounced narrowing of scope. A source sidecar correctly found that Bell Beaker ancestry turnover in Britain does not establish violent invasion, forced removal, or coercive land acquisition, but initially treated that absence as disqualifying. Main adjudication returned to the frozen inclusion rule. A non-state demographic expansion may qualify through directly supported substantial replacement or absorption without a claim of violence. Bell Beaker Britain therefore remains an active expansion analogue, narrowed to the directly supported c. 2450–2000 BCE migration and turnover interval; unresolved social mechanism remains unknown. This does not rescue every ancestry horizon. The combined Lower Danube/Carpathian Yamnaya, Britain/Ireland Neolithic, and continent-scale Corded Ware rows remain held because their evidence does not establish one coherent affected land and population, causal process, or independent profile. Scope admission, mechanism certainty, and event coherence are three separate questions.

Ancient colonial history poses the same event-unit test. Greek Sicily and southern Italy require source-bounded local successors because independent founders, peoples, tracts, and mechanisms do not share one residual profile. The Massalia–Emporion network becomes an inactive parent because real founding and trading connections do not erase separate territorial institutions. The Black Sea and Cyrenaica rows fail closed on evidence holds rather than inheriting continuity from a regional or dynastic label. The already inactive Han Hexi/Western Regions parent is confirmed for the same reason: dynasty-wide continuity cannot merge distinct territories and frontier institutions. These judgments enter the verified register and candidate tests below; they do not individually establish a release-wide fairness result.

The Roman comparison shows what is sufficient to retire a wrapper without pretending that an archive is complete in an absolute sense. The two broad Republican Italy and Po Valley rows are not confirmed from a general chapter title or a long child list. They remain inactive audit rows because a locator-bearing source program dispositions all sixty-seven canonical parent components as bounded successors, nested supports, explicit holds, non-events, or separately reviewed later processes; a ninety-six-unit parent map reconciles the shared children; and a fail-closed validator prevents either parent from coexisting with its replacements. Eleven uncertain foundations remain visible as holds rather than being smuggled into a residual Roman event. Source richness therefore increases the resolution at which claims can be tested, but it does not entitle Rome to extra statistical weight.

Coherent demographic corridors also differ from mere regional periods. The coastal Tupi and Paraná–Plata Guaraní expansions remain separate active units because their source bodies identify distinct settlement streams and connected corridors despite long duration and locally mixed mechanisms. The 1964–present Amazon highway-frontier label and the Highveld Difaqane period remain boundary holds because each aggregates different actors, projects, directions, and land outcomes without a complete successor set. Ancient or modern chronology does not decide any of those four outcomes; source-supported continuity and independent profileability do.

Cases outside the ancient Mediterranean test scope, scale, and historical tempo directly. An uninhabited-island settlement and a post-depopulation occupation cannot be treated alike merely because both lack a resident population at the later settlement date. Bermuda and Cayman remain outside scope because permanent settlement began without a local prior- population land transition. The Bahamas/Turks umbrella remains on

hold because Spanish destruction of Lucayan communities and much later British repopulation are distinct processes whose separation does not yet yield a complete child set. The rule turns on the documented land and population history, not on an island label or a mechanical test of who was present on one date.

The Australian and New Zealand comparisons show that geographic breadth does not itself require either retention or splitting. Western Australia's southwest, Goldfields, Kimberley, and pastoral frontiers remain one explicitly mixed, multiregional event because successive colonial and state institutions connect settlement, policing, labor control, reserves, and missions without a documented reversal. The Queensland-and-wider-north label remains a hold: similar frontier violence appears across it, but one implementing chain and a complete disjoint successor map do not. On the other side of the Tasman, a broad Company-plus-Crown Aotearoa umbrella remains held while the Ngāi Tahu South Island process remains active. Multiple deeds do not force a split when the affected people, land and resource relationship, purchase machinery, Crown responsibility, and continuing grievance supply a coherent profile. Direct land transfer also does not need lethal coercion to qualify.

Algeria and Egypt provide an especially close modern-tempo comparator. The 1961–1963 departure of Algerian Jews remains one event because the affected population, predominant Algeria-to-France route, French-citizenship transition, and concentrated decolonization setting cohere; this does not make every departure individually coerced. Egypt's 1948–1967 umbrella is revised to a required split. The 1948–1951 wartime internment, sequestration, violence, and departure were followed by release and partial normalization; a different regime imposed the 1956–1957 Suez-era expulsions and seizures before partial rescission; and 1967 produced a new wartime detention episode. A shared population and broad Arab–Israeli conflict cannot erase reversals, changing programs, and independently profileable mechanisms. The contrast demonstrates why short or modern chronology receives no automatic preference: continuity retains Algeria, while discontinuity splits Egypt.

The Iran-through-Tunisia comparison shows that boundary correction and claim calibration are not the same operation. Iran and Iraq remain coherent active units, but direct evidence of small, non-reversing returns requires removing absolute “no return” language. That changes the description of the outcome without manufacturing a new event. Libya remains a required split because a completed 1949–1952 mass departure left a separately identifiable remnant that was later forcibly evacuated; Morocco remains one unit because recurring organizations and overlapping cohorts connect the phases without a comparable first-wave closure. Lebanon remains held because the accessible evidence neither joins its crisis phases nor bounds complete children. Syria newly fails closed because the opened passages cover only the late 1987–1993 phase of an asserted 1948–1994 event. These outcomes separate three questions that can otherwise be conflated: whether one event exists, whether its internal claim is phrased accurately, and whether the sources span its whole asserted boundary.

This creates two separate fairness obligations. **Boundary fairness** asks whether each row is one comparably reasoned event and whether any history is counted twice. **Measurement fairness** asks whether its sixty traits were scored with evidence and anchors appropriate to its era, using unknown rather than absence when the record cannot answer. Neither substitutes for the other.

This rule must also operate symmetrically across archives. Rich Roman or Neo-Assyrian records may reveal real local events, but named places and dated inscriptions are not automatic rows. Sparse prehistoric evidence may justify broader uncertainty, but it cannot justify merging demonstrably different processes or converting silence into continuity. Holds and unknown values are the proper response when the evidence cannot support a stable boundary.

The existing register supplies case law and partial diagnostics: retired wrappers, overlap crosswalks, inactive parent rows, era-specific coding anchors, era reweighting, grouped Roman and Hellenistic alternatives, and the Neo-Assyrian archive-density experiment. The reproducible fairness intake contains 689 review units: the 592-

row v4 event/parent intake plus 97 corrected Hellenistic units, of which 92 were model-eligible at intake and five retained for audit only. The earlier 334-cell missingness review and 500-cell fresh double-coding are complete; 374 fresh cells agreed exactly and all 126 disagreements received one evidence-grounded, no-averaging adjudication. That 5,820-cell Hellenistic ledger remains a completed input, not a substitute for checking the boundaries and scores of the rest of the roster.

The release-wide boundary process is now substantially further along. Two model-blind calibration reviewers first supplied independent dispositions for the deliberately difficult calibration set; one no-averaging reconciliation resolved their actual disagreements. A separate full-roster pass supplied the remaining rows. Independent source verification then opened evidence for all 333 preregistered event checks and all 245 relationships, rather than treating a structurally complete spreadsheet as historical verification. Event outcomes were 225 confirmations, eighty revisions, and 28 holds for main adjudication; relationship outcomes were 210 confirmations, nineteen revisions, fourteen holds, and two removals. The verifier recorded 147 findings: 108 event, 35 relationship, and four sample-coverage findings.

One main adjudication reviewed each of those 147 findings once. It accepted 99 bounded verifier revisions and two relationship removals, preserved 42 evidence holds, and recorded four inaccessible scale observations as explicit sample limitations. The resulting non-destructive register preserves every one of the 689 original rows and 245 relationships as an audit-visible prefix. Its effective event states are 518 active retains, 85 boundary holds, 65 required splits, nine inactive audit rows, eight retired active parents, and four scope exclusions. The 356 ordinary retains outside the preregistered source-check sample are labeled as not selected by design rather than being misdescribed as source-verified. Structural validators reproduce the register and confirm that no prospective target or downstream publication authority was silently activated.

The successor gate was kept separate from measurement. It resolved one normalized 233-target admission denominator without treating source support as automatic admission. Of 158 initially eligible targets, 157 cleared the source minimum and one overbroad Somalia target returned to hold. The 157 cleared successors then received two independent sixty-dimension codings and one disagreement-only, no-averaging adjudication, resolving all 9,420 cells.

Thirty accepted boundary findings also affected events remaining active. Seventeen complete profiles passed exact-reuse review; thirteen changed events required 364 freshly coded cells. Their source-complete packets contained 43 source records, and the same two-coder plus disagreement-only procedure resolved every cell. An earlier score-blind pilot lacked consistently openable source bodies and precise locators; its appropriately unknown-heavy returns are preserved as a diagnostic and excluded from measurement. The replacement lane required each non-unknown answer to cite an opened dossier source identifier.

The completed v5 candidate builder applied these dispositions without mutating v4. After the three scope exclusions, it created a 580-event parent reference and a 676-event successor candidate, with all 69 connected histories retained as alternative-resolution groups. Activation required the separate publication decision recorded above.

The source requirement was added after an early process failure: a verifier returned 100 mechanically complete rows but later disclosed that it had opened zero underlying works or passages. The project preserved and disqualified that return rather than treating spreadsheet completeness as historical independence. The replacement run records, for every checked event, the work actually opened, an exact passage locator, an evidence grade, and the source-access state; its completed counts and main dispositions are reported above. Its first checked case remains a useful illustration. Malhi et al. (2008) support a small proto-Apachean movement into the Southwest and extensive admixture, but the opened passages do not establish prior-population displacement,

territorial replacement, or the unity of a c. 1300–1700 event. The source verifier therefore held the broad Athabaskan row, and main preserved it as an inactive evidence hold rather than converting evidence of migration into evidence of one displacement event.

Interim corrected-tranche diagnostic (2026-08-25; not a release result). Adding the 92 corrected model-eligible Hellenistic events yields nine factors under equal-event weighting, whether the four overlapping Greek-colonization parents are retained (675 events) or removed (671). Collapsing linked programs still yields nine; collapsing the entire added tranche to one median restores eight. Across declared weighting and imputation choices, the retained count spans seven to nine. The new events are evidence-sparse on this instrument (median 34.5 of 60 cells missing), and no single event is materially influential under the preregistered leave-one-out thresholds. The defensible interim inference is therefore collective sensitivity to event resolution, corpus composition, and missing-data treatment—not a newly settled ninth trait. The complete 689-row boundary application and verification remain necessary before this can become a release-wide fairness claim.

Historical candidate-boundary diagnostic (2026-08-27; not yet activated at that checkpoint, subsequently promoted as v5). The explicit candidate builder binds every roster decision, score source, parent replacement, overlap group, and date proxy. Three source-resolved scope exclusions reduce the public 583-event roster to a 580-event internal parent reference. The canonical-successor candidate then retires 61 active parents and adds 157 independently scored successors, for `580 − 61 + 157 = 676` events. Nine additional parent references remain audit-visible but were already inactive and are not subtracted twice. Sixty-nine connected histories retain both parent and successor resolutions for direct sensitivity tests.

The model battery treats temporal scale as one diagnostic rather than the event definition. It compares equal events, grouped successor programs, parent-versus-successor resolutions, era balance, observed-duration weighting, source resolution and missingness strata, four imputation treatments, 69 program swaps, 69 leave-program-out runs, and every one of the 676 possible single-event omissions. The corrected-parent reference retains eight factors; the canonical successor retains nine. Grouping each connected program to one median returns eight; so do the complete-parent alternative and removal of all 39 explicit date proxies. Thus chronology is not the event-unit rule, but resolution and provisional dating do materially affect the retained count.

The most important result concerns the factor space as a whole. Under all 676 single-event omissions, the nine-dimensional subspace has a median maximum principal angle of 1.036 degrees, a 95th percentile of 2.433 degrees, and a maximum of 4.808 degrees; no event crosses five degrees. The forced-eight subspace has a median of 1.624 degrees, a 95th percentile of 13.871 degrees, a maximum of 86.489 degrees, and 36 omissions above ten degrees. Across 138 whole-program alternatives, the nine-dimensional maximum angle has a median of 1.810 degrees, a 95th percentile of 4.743 degrees, and a maximum of 5.947; seven scenarios cross five degrees and none crosses ten. Forced eight has a median of 3.345 degrees, a 95th percentile of 73.823 degrees, a maximum of 88.802, and 24 scenarios above ten. No tested individual deletion materially remakes the candidate's nine-dimensional covariance space; coherent program choices can move it modestly and frequently destabilize a forced-eight projection. These diagnostics are consistent with rotation or compression, not proof of one unique cause.

This supports a qualified candidate-level account with three deliberately separate parts. **Procedural boundary evidence** is strong: active rows and relevant inactive parents have explicit dispositions, overlap treatment, evidence trails, and sensitivity groups. **Measurement robustness** is strong for single-row perturbations of the nine-dimensional subspace but sensitive to exact factor count, coherent program resolution, date proxies, weighting, imputation, axis naming, and public-card projection. **Coverage completeness** is not demonstrated and is not claimed: the register is cross-indexed, independently gap-tested, and publicly corrigible under a

declared scope, not a proof that every qualifying event has been found. The final post-v5 Editorial Suite audited 43 substantive changed claim groups and matched all 43, with no overclaim, underclaim, unsupported, or uncheckable verdict inside its declared local-source boundary. Its qualified promotion decision authorizes this bounded fairness wording for v6. The immutable builder and independent release validator passed before the active pointer changed.

The preceding v5 candidate claims are bound to `analysis/second_pass/event_unit_fairness_v1/candidate_models_v2/MODEL_RESULTS_AUDIT_DISPOSITION.md`, `model_results_v2/MODEL_RESULTS_SUMMARY.json`, `subspace_stability_v2/SUBSPACE_STABILITY_SUMMARY.json`, and `factor_retention_calibration_v2/FACTOR_RETENTION_SUMMARY.json`. Their validators and hash manifests accompany each output directory.

The claim sought is deliberately limited: not that historical events have one objectively correct atomic scale or that boundary bias has been eliminated, but that the candidate's boundaries, overlap decisions, and sensitivity groups are published and contestable, and that the measured consequences of declared alternative resolutions are reported rather than hidden.

2.3 Coverage discipline and the gap scan

A register can be honest without being complete, but it must know which it is. Beyond the audit's country sweep, a six-angle adversarial gap scan (archaeogenetics; ancient/medieval non-Western; development and conservation displacement; modern conflict and partition; Oceania and island worlds; internal colonization in Russia, Asia, and the Americas) proposed forty-five candidates, each then adversarially verified against the register with rejection as the default. Thirty were confirmed in the initial verification and fifteen were held unverified at a strength cap; three of the held candidates were subsequently verified and confirmed, for **thirty-three confirmed additions with twelve candidates still recorded as unverified**. Among the confirmed: the class-based Soviet dekulakization system (Viola 2007; Polian 2004), which the register's ethnic-deportation coverage had missed entirely; the Porfirian Yaqui deportations (Hu-DeHart 1974), the register's first event with Mexico as perpetrator; the Argentine Chaco conquest (Gordillo 2004), a gap the register's own Argentina entry had named; the Neo-Assyrian and Neo-Babylonian deportation systems; Umayyad Iberia; the Danelaw; Ottoman sürgün; and a cluster of dam, park, and military removals (Kariba, Volta, Narmada, Kaptai, Serengeti/Ngorongoro, Bwindi, Vieques). The twelve still-unverified candidates remain recorded as such rather than silently dropped (`analysis/gapscan/gapscan_results.json`).

The due-diligence protocol formalizes this stance (`audit/COVERAGE_DUE_DILIGENCE.md`): freeze the inclusion scope before searching; give every branch and lead an explicit row-level disposition (advance / existing-event support / duplicate / evidence hold / searched with no bounded event found / inaccessible source / unresolved); and preserve every precise fact, contradiction, and negative finding with a document locator, whether or not it enters atlas prose. "Searched and unsupported" must remain distinguishable from "never examined." The register does not claim worldwide saturation; it claims that its blind spots are enumerated.

This is selection for qualifying positive cases, not a representative sample of taking versus coexistence. More included cases can document recurrence but cannot supply the missing denominator for universal propensity, inevitability, relative prevalence, or ultimate causal claims.

2.4 The ancient and premodern expansion

The first modeled corpus (407 events) inherited the audit's modern-forward center of gravity. Because the atlas's editorial objective is to demonstrate documentable patterns *across millennia*, the second pass ran bounded source-tranche expansions: nine initial ancient/medieval cases, five Achaemenid, four early Sasanian, two New Kingdom Egyptian, nine Mongol western-campaign cases, the Neo-Assyrian replacement and expansion described above, and finally the ninety-five-event Roman tranche — each tranche passing a per-event source brief, an event-unit gate, an overlap crosswalk, and full sixty-dimension coding before activation. (A further Hellenistic-eastern pre-coding queue, anchored in the standard settlement catalogues — Cohen 2013 for the East — was held outside that release pending the same gates.) The current statistical release, `second_pass_v6_675`, holds 675 active events in an immutable, hash-verified directory. It preserves v5, v4, and the prior v3, 490-event, and 407-event checkpoints. V6 applies the finite post-v5 retirement, cell, chronology, family, and public-narrative corrections without silently editing the old release. Two active rows retain explicit family-classification evidence holds and remain outside atlas rendering.

2.5 The source inventory

Every event carries at least one directly relevant source; recommended atlas additions carry at least two. The census for this monograph — regenerated by the saved script `monograph/build_bibliography_census.py` — finds **854 unique URL records** across the audit registry, the legacy atlas registry, the gap scan, and the second-pass tranche ledgers (`monograph/bibliography_census.csv`; 845 at the fact-check checkpoint — the delta is seven audit-register sources added with the Sobaipuri and Salinas admissions plus two sources from the Apuan/Luna recode file, whose omission from the generator's harvest was a coverage gap found in the second review). This is a *source inventory*, not yet a publication bibliography: it deduplicates by normalized URL only, so it contains generic slot titles from the Roman tranche, primary-source URLs alongside secondary works, 23 duplicate exact-title groups, and at least one confirmed same-work-at-two-URLs duplicate. Separately, the Greco-Roman disposition ledger holds 398 rows with a primary-source locator (classical texts via the Perseus and Thayer corpora: Livy, Diodorus, Appian, Arrian, among others), and the early Neo-Assyrian ledger holds 75 rows drawing on 33 distinct royal-inscription source identifiers (pipe-split rule; verified). A work-level scaffold now exists (`monograph/build_work_level_bibliography.py`): the 854 URL records resolve to 852 conservative pre-review groups. All fourteen near-title candidates have explicit dispositions in `bibliography_review_queue.csv` — twelve different works, one same work, and one chapter-within-book relationship. Applying the one reviewed merge produces **851 work groups**: 544 titled works, 155 primary-source loci, 122 DOI-keyed works, and 30 modern slot records. The generator preserves decisions on regeneration and never merges `different` or `part_of` rows. Publication form is decided (2026-08-21): the anchor list remains in the monograph and the 851-work inventory publishes as a separate versioned dataset with a methods preface. Typed bibliographic metadata remains the production step before that dataset release.

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3. The instrument

3.1 Five batteries, deliberately unmerged

To measure events without adopting any one framework as the sole authority, five disciplinary design briefs were commissioned independently — settler-colonial studies; political economy of land and labor; historical demography and archaeogenetics; genocide and forced-migration studies; state formation, sovereignty, and

ideology — each producing twelve candidate dimensions with concrete 0–4 anchors, evidence notes, era-coverage rules, and *a-priori predictions of the factor structure*. Those predictions were **documented in a dated local file before the surviving coding outputs** (`analysis/codebook/a_priori_predictions.md`); they were not preregistered in the formal sense of a time-stamped, read-only external deposit, and we do not claim that status for them. The five batteries were then adopted **verbatim and side by side**: sixty dimensions, no merging, no adjudication of overlap. Where two disciplines measure nearly the same thing (settler-colonial "demographic saturation," demographic "incomer share at consolidation"), the overlap is retained as data — convergence between independently designed instruments is a result, and hand-collapsing it in advance would have pre-empted the factor analysis that §5 reports. Retaining overlap is itself a theoretical design choice: repeated indicators can give some constructs more influence in the covariance model. [Battery and anchors: `analysis/codebook/codebook.md`, v1.1.]

3.2 Anchored ordinal scales

Each dimension is scored 0–4 against written anchors chosen to be decidable from evidence in minutes of research, not vibes ("settlers exceeded 50% of the post-event population within two generations," not "significant settlement"). Anchors carry era-coverage rules: a prehistoric case scores sex-biased admixture from ancient-DNA evidence (e.g., steppe-ancestry turnovers: Haak et al. 2015; Olalde et al. 2019) where a modern case scores passenger manifests and censuses.

3.3 Unknown is not inapplicable

The pilot (§4.1) surfaced a distinction that became one of the instrument's most productive features. `−1` (*unknown*) means the dimension has a real referent but research cannot settle the score — native mortality in a prehistoric expansion. `−2` (*structurally inapplicable*) means the referent does not exist — incoming-settler dimensions for a pure expulsion; prior-population dimensions for settlement of uninhabited land (Norse Greenland). The two are different facts about the world, and the pattern of `−2` codes is itself structural signal: events with no incomers to describe are a substantial fraction of the corpus, and no continuum connects them to family-farm colonization. Treating that as "missing data" would have erased the register's clearest structural boundary. The v3 release types every one of its 34,980 cells as a numeric judgment, an explicit unknown, or a structural not-applicable, with zero untyped blanks.

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4. Measurement with AI research agents

At the historical v3 checkpoint, this project's coding labor — sixty dimensions × 583 events, with per-event source research — would have been impracticable within this single-scholar project's available time and funding. It was performed by large-language-model research agents. We describe the protocol in the same spirit as any instrument: what was tested, what failed, what the uncertainty diagnostics are, and what cannot currently be recomputed. Two framing cautions govern the whole section: everything below measures *agreement between coders*, which bounds reliability but is not evidence of historical validity against an expert gold standard; and the process was iterative, with the instrument itself amended once (the `−1`/`−2` rule) after the pilot. (Cf. emerging practice on LLM annotation in social science: Gilardi, Alizadeh, and Kubli 2023; Ziems et al. 2024.)

4.1 Pilot

Two independent agent coders (separate contexts, no communication) coded the same six maximally different events — Yamnaya lower Danube, Greek Sicily, British New England, the Crimean Tatar deportation, Chagos, Norse Greenland — on all sixty dimensions with per-event web research. Agreement: **0.93 exact, 0.99 within one anchor step, mean absolute difference 0.09** on the 0–4 scales, over **345 scored comparisons** (unknown-involved pairs excluded). The pilot is an instrument-debugging exercise at small scale, not evidence of accuracy; and because both coders independently reported the same defect (no way to say "no referent"), the `−1`/`−2` distinction postdates the pilot and was **not** validated by it (`analysis/coding/pilot/agreement.json`).

4.2 A model-selection gate that failed a model

Before production, a cheaper candidate coder model was trialed against a banked 30-event reference set (1,564 numeric cells; 236 N/A-involved cells). It scored **0.67 exact**, mean absolute difference 0.45 — and, decisively, **0.22 concordance** on the `−1`/`−2` distinction, which would have corrupted precisely the missingness structure §3.3 depends on. The candidate was excluded; the full corpus was coded on the stronger model, at material extra cost. We report this because instrument failure that is caught and documented is evidence the gate works, and because "which model coded your data" is as reportable as "which assay kit" (`analysis/validation/agreement_sonnet.json`).

4.3 Production protocol

Events were dealt round-robin into era- and mechanism-stratified batches of about ten, so no coder anchored on a single event type. Each coder read the sixty-anchor codebook and a per-event research brief (register summary, actors, dates, audit critique, vetted sources), performed bounded web research, and returned schema-enforced integer vectors with per-dimension flags and consulted-source URLs. Prompts included an anti-central-tendency instruction (score the anchor the evidence meets, not the case's reputation) and, after §4.2, worked examples of the `−1`/`−2` boundary.

4.4 Cross-model agreement, its weak point, and a recomputation gap

Thirty events were independently coded by two different production models: **0.76 exact, 0.95 within-1, mean absolute difference 0.31** over 1,704 numeric comparisons — the observed cross-model agreement estimate for this stratified 30-event sample, well below the same-model pilot. On the 96 cells where either coder used a non-numeric code, however, concordance on *which* non-numeric code was only **0.50**: the models agreed much better about scores than about the boundary between "unknown" and "inapplicable," and downstream analyses of missingness patterns inherit that uncertainty. Measurement error at these levels may attenuate or distort the recovered structure; because model coding errors can be systematic rather than random, we do not claim the true structure is sharper than the measured one. A reproducibility gap flagged by the v0 fact-check is now closed: the raw coder journals behind both model comparisons were recovered from the drafting session's working directories, copied into the project at `analysis/validation/raw/` with SHA-256 sums, and both stored comparisons recompute exactly from the in-project copies (independently reproduced in the second review, and re-verified here: sonnet 0.669 exact / 0.216 N/A concordance; cross-model 0.763 / 0.953 / 0.500).

A clean-data stage then repaired first-pass defects without touching the raw record: thirteen duplicate submissions reconciled by rule (agreement kept; means for numeric disagreement; manual adjudication for structural disagreements, each documented); two genuinely missing cells resolved by explicit adjudication; gap-

scan year fields completed. The repaired baseline is frozen and hashed (`analysis/second_pass/baseline_407/`, `data/duplicate_reconciliation.csv`).

4.5 The human boundary, and what the record actually is

The documented consequential decisions reviewed to date — event universe, severity-axis inclusion, research depth, the no-merging rule, family adoptions, case retirements, release promotions — were reserved for or confirmed by the human author in the project's decision ledgers. The bounded transcript audit subsequently corroborated every major decision it checked; this is not a claim that every minor action across the project has been exhaustively traced. The agents designed instruments *within* those decisions, coded *under* them, and analyzed *for* them.

A standing project rule requires every precise source-backed fact found during research to be written into the durable file record with a document locator, because working-session context is unreliable as an archive. The durable record is broader than prose: dated Markdown reviews, CSV ledgers, JSON results and manifests, scripts, and hash-verified releases. During drafting, v0 claimed the second-pass working conversations were unrecoverable after two transcript-store searches returned nothing; a subsequent audit found **five surviving session files** (16 and 19 August 2026) containing substantive second-pass discussion. The claim is reversed here, and the bounded recovery pass has since been **completed** (`analysis/second_pass/PROCESS_FACT_RECOVERY.md`): six process facts newly recorded (among them the provenance of the reconstructed Neo-Assyrian source series and an unlinked report deployment), twelve major decisions confirmed with session-and-line locators — including the decision stream reserved to the human author, which the recovered record corroborates for every decision checked — and explicit negative findings for the sessions containing nothing unique. The episode is retained as a worked example of why the preservation rule exists: neither chat *nor a single failed search of chat* is a substitute for the written ledger.

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5. Results I: a persistent trait structure with a boundary-sensitive count

5.1 The factor solution at the analysis baseline

Parallel analysis (Horn 1965) against a simulated null retains **eight factors** on the 407-event first-pass matrix and again on the 490-event expansion. The axes of that eight-factor solution, in descending variance:

1. **Government direction** (state orchestration versus settler agency) — organizing-agent locus, carrier centralization, perpetrator state centrality, metropole anchorage, intent legibility.
2. **Legal and property machinery** — title-transfer formalization, formal expropriation, land commodification, legal instrumentation, cadastral legibility.
3. **Settler implantation** — family-complete migrant streams, permanence orientation, stream coordination, self-reproduction, saturation.
4. **Elimination versus incorporation** — prior-population removal and ancestry non-persistence versus native-labor incorporation.
5. **Lethal coercion** — lethality, mortality share of decline, subsistence destruction, survivor confinement, sex-differential targeting.

6. **Durability of outcome** — sovereignty durability, structural persistence, present-day endpoint, return foreclosure.
7. **Displacement into occupied space** (statistical form; see §5.3).
8. **Coerced outside labor** — the exploratory factor mixes unfree incomer share and exogenous-labor structure with household, distance, and subsistence correlates; the corrected public construct is described below.

At 407 events these carry 53% of variance on sixty dimensions with none dropped for missingness. Dimension-level convergence across disciplines is pervasive: the first factor alone draws its top loadings from four of the five batteries — the independent instruments agree about what co-varies, which is precisely what the unmerged design was built to reveal.

5.2 What growth does to the count

[TECH] The factor *space* persists as the corpus grows; the factor *count* does not:

Checkpoint	Parallel-analysis retention	Ninth-factor margin
Repaired 407	8	—
Earlier 490	8	−0.045
Earlier v3 583	**9**	**+0.019**
Earlier v4 583	**8**	**−0.00789**

Between 407 and 490 events, per-axis loading congruence runs 0.896–0.985 and shared-event factor-score rank correlations 0.934–0.988. In v3's 583-event roster, the default analysis crossed the retention threshold for a ninth factor by a margin of +0.019 — a boundary result, sensitive to missing-data treatment (complete- case subsets retain eight; subsets restricted to well-documented events retain nine) and to era and source weighting. Forcing eight factors at 583 recovers the named subspace but rotates several axes: individual 407→583 congruences range **0.653–0.958**, with the durability, legal-machinery, and coerced-labor axes least stable. The ninth pattern itself is historically intelligible but impure: it contrasts well-documented, mostly modern systems of territorial closure, administrative classification, and codified legitimation with source-sparse, mostly ancient cases, and it correlates strongly with both chronology ($\rho = +0.73$ with start year) and documentation density ($\rho = -0.68$ with unknown-cell count). "Documented territorial-administrative closure" is a fair description of what it measures; it splits and rotates a correlated block rather than adding one independent trait, and it is not robust enough to become a ninth public bar (`ROMAN_FACTOR_COUNT_SENSITIVITY.md`). The honest summary is not "eight factors, replicated," but: **a reproducible multidimensional covariance structure whose exact factor count sits on a documented boundary between eight and nine, entangled at that boundary with how much the sources let us know.** V4 retained the same event count after bounded substitutions but returned to eight, with a ninth-factor margin of −0.00789.

The corrected 676-event successor candidate, later frozen as v5, sharpened that conclusion. Its Horn result was nine in every one of twenty independent 500-draw calibrations and under every single-event deletion. Yet grouping connected successor programs, using the complete-parent alternative, collapsing all successors to one median, or dropping the 39 date-proxy rows returns eight. Weighting and imputation alternatives are material as

well. The ninth direction is therefore a reproducible feature of the canonical candidate under its declared preprocessing, not a universal ninth trait.

Whole-space perturbation resolves what per-axis congruence alone could not. Across all 676 leave-one-event runs in that v5 battery, the nine-dimensional space never moves more than 4.808 degrees, whereas forced eight reaches 86.489 degrees and crosses ten degrees 36 times. Whole-program alternatives keep the nine-dimensional maximum below ten degrees but show modest movement beyond five degrees in seven of 138 scenarios. The public question is therefore not simply “eight or nine bars?” The current eight-bar card remains an explicitly editorial, evidence-aware projection for this release; the candidate result does not mechanically add a ninth bar.

The finite post-v5 correction cycle provides an additional bounded post-v5 perturbation check of that interpretation. One wrapper retirement plus 117 source-adjudicated cell changes produced the 675-event candidate subsequently frozen as active v6, which still retains nine factors. Its nine-dimensional space is 2.765 degrees from immutable v5, with minimum principal-angle cosine `0.998836`; forced eight moves 6.685 degrees and gives misleadingly low same-name comparisons for three axes even though best-matched eight-factor loadings remain highly congruent. The correction itself is not the source of the broader sensitivity: equal-era weighting and the three declared alternative imputation treatments move the corrected nine-factor space by 26.987, 32.608, 25.249, and 70.014 degrees, closely matching the v5 values of 26.539, 36.124, 24.977, and 71.313. The candidate therefore preserves both findings at once: source-adjudicated roster and cell corrections do not overturn the canonical structure, and the exact factor allocation remains conditional on how a source-uneven corpus is weighted and completed. The 70.014-degree structural-zero alternative is an extreme diagnostic bound, not a preferred completion model.

These v6 promotion results are bound directly to the preserved candidate outputs `analysis/second_pass/v5/post_v5_correction_application_v1/candidate/post_v5_corrected_successor_675/results/post_v5_model_comparison_v1/MODEL_COMPARISON_SUMMARY.json` and `analysis/second_pass/v5/post_v5_correction_application_v1/MODEL_COMPARISON_INTERPRETATION.md`. The old 583/679 output is retained separately as a superseded diagnostic.

Two further caveats from the v3 stability battery. Durability is the one era-confounded axis (Spearman $\rho = -0.51$ with start year): older events have had longer to look permanent. Lethal coercion, by contrast, is era-independent in that battery ($\rho = 0.00$), and on-atlas versus off-atlas events differ in mean lethality by $+0.03$. This supports keeping severity separate from category labels; it does not validate the normalized mechanism taxonomy, which separately failed its reliability gate (§6.4).

5.3 From factors to the public card: a face-validity correction

The public eight-trait case card is **not** the exploratory factor solution, and after v3 it deliberately is not: six bars are transparent weighted composites of coded dimensions that load consistently in *both* the 407 and 490 snapshots; the seventh public bar replaced the statistical seventh factor after manual Roman review exposed a face-validity failure (a case whose residents demonstrably stayed could receive a high "displacement" bar, because the factor mixed relocation with reversed land-conversion dimensions). The public seventh bar is now a direct, evidence-gated **Separation from homeland** measure (relocation distance + return foreclosure, suppressed when no population was displaced), with the statistical factor retained for diagnosis (`EIGHT_TRAIT_CARD_SCORING.md`). The card survives the eight/nine boundary in §5.2 precisely because it never depended on the factor count: it is an editorial summary of coded judgments, defended as such. We flag this as the paper's model case of the intended division of labor: statistics propose, source-anchored review disposes.

Public F8 face-validity correction (2026-08-27). Atlas authoring exposed a second instance of the same general problem. The factor-informed bar labelled “Coerced outside labor” included three conceptually secondary correlates— household composition, geopolitical distance, and reversed subsistence destruction—alongside the two dimensions closest to imported or compelled incomers. Events with no imported labor stream could therefore receive nonzero, apparently well-supported values.

The corrected public F8 is **Imported unfree labor**: whether a distinct population was brought from elsewhere as enslaved, indentured, or otherwise unfree labor. `scs_exogenous_labor_triad` is the applicability gate; when the construct exists, `pel_unfree_incomer_share` helps scale it. Unknown applicability remains unknown, structural N/A suppresses the bar, and a known zero gate displays zero. One bounded exception permits applicability when an event is itself the importation of unfree labor but the statistical triad is N/A only because no settler population exists.

The frozen v2 comparison changes 540 of 676 public cards by at least five points and 226 by at least twenty; 450 old nonzero readings become direct zeros and 58 cases are structurally inapplicable after review. Eight gate conflicts were individually adjudicated with sources, leaving zero unresolved. The complete 224-case/234-frame implementation then passed one predeclared desktop and mobile audit with zero issues. This magnitude rules out a quiet case-specific patch and justifies a corpus-wide public correction. The statistical F8 remains an exploratory covariance axis in the immutable release; no factor score was rewritten. The generator, comparison, review queue, model, implementation disposition, and hashes are preserved in `analysis/second_pass/v5/eight_trait_card_f8_review/`.

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6. Results II: no tested partition survives as taxonomy

6.1 The clustering looks fine until you test it

K-means on the eight-axis scores yields silhouette-optimal solutions (k = 10 at silhouette 0.24 on the first pass) that pass casual inspection. Testing undoes them. Bootstrap resampling (Hennig 2007) finds only four of ten clusters stable; the stability-optimal k = 6 solution (mean Jaccard 0.842 on the first pass; 0.803 at 490 events) is *stable but shallow*: absolute silhouettes near 0.24 sit far above matched-covariance unimodal noise (26–32 standard deviations — genuine density structure exists), yet five hundred restarts produce 367 distinct near-tied partitions, and growing the corpus from 446 to 490 reviewed events moves 71 of 446 shared cases between free-solution neighborhoods (ARI 0.661). Carrying the prior partition forward costs less than 1% in within-cluster error — many assignments are exchangeable among the tested near-ties even though the density structure is not.

We state the conclusion at its defensible strength. These diagnostics evaluate *this encoding, these algorithms, and this corpus*; clustering cannot prove that history contains no natural kinds, and stable clusters can themselves be artifacts of method (Hennig 2015). What the record shows is narrower and sufficient: **no tested automatic partition is robust enough to serve as the atlas's public taxonomy.** The phrase "no natural species," where it appears in project material, is an editorial metaphor for that result, not a claim about historical ontology.

6.2 The archive-density experiment

[TECH] Why do partitions move? A controlled sensitivity replaced the forty-four early Neo-Assyrian events — separately valid, separately bounded, but drawn from one unusually granular archive — with non-historical

medians at three resolutions. One median per ruler (three rows): partition agreement with the 446-event baseline rises from ARI 0.661 to 0.928 while minimum factor congruence stays ≥0.999. One median per coding pattern (nine rows): 0.968. The factor axes barely notice; the clusters reorganize around documentation density. The scoped inference: **in this corpus and this design, automatic clusters weighted history by its archives** — a well-recorded decade of Assyrian campaigning outvoted a poorly recorded century elsewhere, not because more happened but because more survives per event. We conjecture the mechanism generalizes to other historical clustering exercises, and the diagnostic (resolution-matched median replacement) transfers directly (`19_early_neo_assyrian_resolution_sensitivity.py`; `SECOND_PASS_INTERIM_RESULTS.md`).

6.3 What the old categories look like from here

Agreement between data-driven partitions and the register's twelve mechanism families is ARI ≈ 0.29 — real, partial, and uneven. Three families are structurally coherent: population exchange/partition (one neighborhood), conflict displacement (97% concentrated), demographic expansion (96% concentrated — the prehistoric cases long hedged as "analogues" are among the *tightest* groups in the corpus, and the hedge understates them). Two families fragment across five neighborhoods each — "state resettlement/internal colonization" (largest share 33%) and "expulsion/deportation/ethnic cleansing" (35%): names for intent or legal character, applied to processes with fundamentally different architectures. The middle of the vocabulary ("conquest with settlement," "genocide-linked clearance," "enslavement/labor removal," "development displacement") is mixed.

6.4 The resolution: families as reviewed filing, traits as measurement

The editorial consequence is not "abandon categories" — a public atlas needs findable groups — but a demotion of their epistemic status. The atlas adopts **six outcome families as a manually reviewed filing system**, informed by the measurements and adjudicated case by case in a written ledger (146 rows at the 490-event checkpoint; every disagreement between proposed label and nearest statistical prototype resolved by source-anchored review, with the source-first outcome retained in all nine early-Assyrian disagreements):

The following table preserves the historical v5 snapshot (2026-08-27), not current v6 family totals. Frozen v6 has 673 assigned events and two family evidence holds among 675 modeled events, as stated in the opening snapshot.

Family (historical v5 labels)	Events (of 676)
Expulsion or flight without replacement	182
Conquest and incorporation	167
State-directed demographic remaking	144
Settler replacement	111
Low-state durable expansion	45
Land clearance for state or project use	25
Family evidence hold	2

Each rendered case additionally carries the **eight-trait profile** of §5.3 — evidence-aware 0–100 summaries of the underlying anchored judgments, with evidence-coverage values and conservative unknown ranges, and bars suppressed where structurally inapplicable. For this release the public card presents one reviewed family and eight measured lenses; normalized mechanism tags remain an internal research layer after failing the reliability gate below. The card is "a set of interpretable lenses, not a claim that history contains exactly eight natural kinds."

Mechanism/family diagnostic (candidate 676; terminal 2026-08-27). The normalized test is reproducible, but its binding disposition after the single permitted repair and retest is `FAIL_RELIABILITY_GATE_MECHANISMS_INTERNAL`. The source-based, family/model-blind mechanism review produced 668 tagged events and eight holds; the independently reviewed family roster has 674 labels and two nonoverlapping holds. The pre-repair diagnostic therefore contains 666 events.

The first run was not accepted silently. Its one permitted final adversarial review found that named dense research series crossed folds, that the program cap omitted 43 early Neo-Assyrian cases, and that declared clustered uncertainty and predictive-stratum outputs were absent. The failed run and audit remain preserved. The corrected version binds an explicit 676-event series crosswalk, joins linked event-unit and dense-series components before fold assignment, caps each named series at five, and adds the missing outputs. It contains 410 unified groups; the corrected program-capped analysis contains 506 events. The full battery and structural validator reproduce exactly with zero group leakage.

In the pre-repair diagnostic, primary mechanism and family are strongly related (Cramer's V = 0.637; unified-group bootstrap interval 0.603–0.681), but they are not interchangeable: normalized residual family entropy is 0.616, with a unified-group bootstrap interval of 0.556–0.638. Primary mechanism is recorded as `coequal_compound` for 186 of 666 events, so this descriptive table is not a dependence analysis of the complete multi-label tag sets.

Repeated unified-group five-fold prediction gives macro-F1 0.684 from mechanism tags, 0.614 from the public eight traits, and 0.771 from context, traits, and tags. Adding tags to context plus traits improves held-out log loss by 0.125 on average—about 11.6%—with a unified-group bootstrap interval of 0.062–0.201; the corrected 506-event program cap preserves the same direction. Conversely, family alone only weakly reconstructs the full

multi-label mechanism sets (macro-F1 0.429, micro-F1 0.620). These results support the conceptual distinction used here: mechanism describes *how* an event operated, while family summarizes its dominant structural outcome. But those estimates were fitted to the first adjudicated mechanism roster. They do not establish that the mechanism vocabulary can be applied independently and consistently enough to support a public layer.

The first independent mechanism review had already missed the frozen reliability floor (micro kappa 0.686, positive-agreement Dice 0.749, tag Jaccard 0.599). The project therefore used its one authorized repair cycle: 269 discordant or diagnostic events received source-first review; all 246 nonautomatic comparisons were adjudicated without averaging; the existing fourteen tags were clarified rather than renamed or expanded; and all 676 events were reassessed under the clarified rules. The repaired roster contains 527 final or evidence-limited tag sets and 149 explicit holds—109 for defective or mixed event units, 31 for evidence, and nine for incompatible complete tag readings. Held rows were not converted into all-negative assignments.

The single permitted retest was then frozen before the reviewer began. Its 160-event sample covered all fourteen tags, all six families, 41 nonempty family-by-era cells, sixteen macroregions, nine dense research series, retained and successor events, on- and off-atlas events, all evidence bands, changed cases, holds, and unchanged controls. The reviewer saw source evidence and the repaired codebook but not the repaired assignments, families, traits, models, sampling ledger, editorial summaries, or prior returns. Coverage cleared the anti-attrition gate: 108 events were jointly reviewable, above the minimum of 75 and equal to 83.1% of the 130 repaired-reviewable sample events.

Reliability did not clear the predeclared gate. Micro Cohen's kappa was 0.732, positive-label Dice 0.790, and tag Jaccard 0.653, versus a required 0.80 on all three. Nine common tags fell below the tag-specific kappa floor, and saved stratum screens found systematic aggregate or tag-level reversals. Exact tag sets agreed for only 28 jointly reviewable events; the frozen disagreement queue contains 121 events. These are pre-adjudication scores and are not altered by later interpretation.

The binding result is therefore `FAIL_RELIABILITY_GATE_MECHANISMS_INTERNAL`. This is not evidence that mechanism and family mean the same thing; it is evidence that this mechanism instrument is not reliable enough to bear a second public analytical label. For this release, the atlas should keep one primary outcome family and the evidence-aware trait bars, preserve source-grounded process prose, and not add normalized mechanism chips to the card. Legacy free-text mechanism labels should be retired from analytical prominence during the next UI cleanup, while their wording and the complete repaired ledger remain available as provenance. The model battery is not rerun on the repaired tags, and the six provisional primary-mechanism summaries are not release blockers.

The stopping rule was frozen before the 269 source-first returns were opened: one repair, one fresh blinded retest, and no recursive tuning. Failure therefore does not trigger another repair cycle. A future public mechanism layer would require new evidence, a materially redesigned instrument, or a separately authorized later release. The corrected 580-parent, grouped-parent, and 69 parent/successor mechanism sensitivities remain unestimated rather than being filled with legacy labels from unreviewed wrappers. The binding output is `analysis/second_pass/mechanism_family_v1/candidate_676_analysis_v2/source_first_review_v1/repair_v1/application/retest_v1/comparison/MECHANISM_RETEST_COMPARISON_SUMMARY.json`; the frozen decision rule is in `analysis/second_pass/mechanism_family_v1/CANDIDATE_676_ANALYSIS_IMPLEMENTATION_SPEC_V2.md`. The superseded run and its audit remain in `candidate_676_analysis_v1/`.

Historical normalization checkpoint (2026-08-27; superseded by the reviewed diagnostic above). The initial 679-event proposal is preserved but superseded by the scope correction described above. The same family-blind normalization now gives all 676 corrected candidate events fourteen composable broad process tags with

no untagged rows. It preserves all 331 raw mechanism strings, flags all 676 event assignments for evidence review, and treats both the 320 bespoke narratives and the inherited labels as starting proposals rather than final data. The inherited labels require review because labels such as “conflict displacement,” “conquest with settlement,” and “development/conservation displacement” combine context, actions, and outcomes that the final tags must separate.

On the 519 corrected-candidate events whose reviewed v4 family remains available, a provisional ordinary five-fold model predicts family at 58.4% accuracy from the proposed tags, 78.0% from the public eight traits, and 86.1% from tags plus traits, against a 27.4% majority baseline. The incremental result was large enough that public redundancy could not be assumed, but it was not dispositive because the tags were unreviewed and the folds did not hold linked programs and dense source series together. The reviewed, audit-corrected grouped analysis above supersedes this checkpoint without erasing it from the audit trail.

The v5 successor-family gate closed. Four source-only primary reviews covered all 157 successors; a separately blinded 45-event audit agreed on 37 and sent eight to one no-averaging adjudication. The resulting full candidate roster contained 674 analysis-ready family labels and two evidence holds. Four disjoint, source-only mechanism reviews then covered the corrected 676-event roster without access to family labels, traits, or model outputs. A bounded 76-event audit and one disagreement-only adjudication produced the first internal roster. The later bounded repair and blinded retest did not clear the public reliability gate. The current public direction is therefore one outcome family plus the evidence-aware trait bars, without normalized “How it happened” chips in this release.

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7. Results III: the atlas as an editorial sample

Measuring the whole register, including everything the atlas excludes, makes the atlas's own boundary legible. At the August 2026 measurement baseline (222 rendered cases), on-atlas events differed from off-atlas events most strongly in settler implantation (+0.99 SD), legal machinery (+0.66), and durability (+0.52), and negatively in government direction (−0.59) and displacement into occupied space (−0.59) — with, again, no lethality difference. This is a coherent implicit definition of the atlas's historical subject, now stated explicitly rather than enforced invisibly. Two large, internally consistent regions of the space were nearly absent (transient conflict displacement: 4 of 66; state demographic engineering: 17 of 61); one measured axis — coerced outside labor — was *invisible* when the analysis was restricted to on-atlas cases only (loading congruence 0.06), a pattern consistent with the atlas's selection having excluded most of the plantation-configuration variance. Whether to move the boundary is an editorial question; the contribution of the measurement is that it is now a *visible, quantified* question, answered deliberately in a 587-row implementation ledger that partitions every release row into retain, review, replace, author, hold, or exclude (`ATLAS_V3_INTEGRATION.md`).

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8. From analysis to editorial practice

A study that ended with a statistical release would misrepresent the project; half its substance is the machinery that turns analysis into a public artifact without laundering uncertainty. Concretely:

- **Versioned immutable releases.** Statistical corpora are promoted once, hash-manifested, and never edited. The active release is `releases/second_pass_v6_675/`; v5, v4, v3, the 490-event, and 407-event

baselines remain preserved. Statistical inclusion is explicitly *not* publication approval; cautioned rows travel with their cautions attached.

- **Retirement with provenance, and successors that earn admission.** Cases that fail review are retired, not deleted: retired umbrellas remain as inactive audit rows that must never render beside their bounded children; Norse Greenland's stable link redirects to the qualifying Thule case. The broad Athabaskan migration was retired with an explicit notice, and its bounded successors — the 1762 Sobaipuri-O'odham relocation, the 1667–1672 Las Humanas abandonment, and the 1676–1677 Quarai/Chililí-to-Tajique relocation — then passed the full admission pipeline (source brief, row-level dispositions, overlap crosswalk, event-unit gate, family review, map-authored routes and land outcomes) and were published in atlas build `20260828z` (then 313 cases, 323 playback stops). Their sixty-dimension coding and v4 statistical admission are complete.
- **One canonical playthrough.** There is no "contextual" second-class layer; an admitted case is a case. (Boundary decisions are therefore real decisions.)
- **A correction channel inside the atlas.** Case-scoped and whole-atlas reports enter the same candidate/correction ledger as the audit's own findings; public criticism is treated as a discovery mechanism, not a threat. The project calls this its Cunningham's-Law posture — the fastest route to a right answer in public is a confidently wrong one — and adopts it deliberately: a strong provisional register with a working correction channel, rather than an expert-review gate before launch.
- **Cartographic honesty rules.** Markers must visibly sit on land; only routes may cross water; a label that cannot be placed truthfully is omitted ("false spatial precision is worse than omission") — the map-level analogue of the register's evidence discipline (`MAP_RENDERING_RULES.md`).

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9. Limitations

1. **Agreement is not validity.** Every number in §4 measures coder agreement; none measures correspondence with an expert gold standard. The cross-model bound is 0.76 exact — and only 0.50 on the unknown-versus-inapplicable boundary — so individual cell values should never be quoted as facts without their sources, and missingness-based analyses carry extra uncertainty.
2. **The factor count is a boundary result.** Eight at 407 and 490 events, nine in v3's 583-event roster by +0.019, eight in the v4 583-event roster, nine in v5's 676-event release, and nine in the active v6 675-event release under their declared preprocessing (§5.2). Grouping, date-proxy, weighting, and imputation alternatives move the count or factor space. Any presentation that fixes either eight or nine as a discovered constant misstates the record; the eight-bar card is editorial. The active 675-event v6 successor reproduces nine factors and a 2.765-degree nine-dimensional distance from v5, but its forced-eight comparison moves 6.685 degrees; that contrast is why whole-subspace diagnostics accompany named-axis comparisons.
3. **Validation raw inputs were nearly lost.** The coder journals behind the model gate and cross-model check lived only in session working directories until the fact-check flagged the gap; they are now copied into `analysis/validation/raw/` and both comparisons recompute exactly (§4.4). The episode argues for depositing raw instrument outputs at generation time, not at audit time.
4. **Model-generated coding.** The instrument is anchored and agreement-tested, but it is not an expert panel; specialist review — invited through the correction channel — will find individual codes to fight about. The project's position is that a documented, contestable code beats an undocumented intuition.
5. **The corpus inherits its archives.** §6.2 is a warning about clustering, but it applies to everything: the 675 frozen v6 events sample what survives. The disposition ledgers (advance / hold / searched-with-no-

finding / inaccessible) bound, but do not remove, survivorship. This selected-positive-case corpus lacks a representative denominator for universal human propensity, inevitability, or the prevalence of peaceful alternatives (§2.3).

6. **One axis is era-confounded** (durability, $\rho = -0.51$ with start year); read old events' durability bars accordingly.
7. **The work grouping is not yet a finished bibliography.** The 854 URL records now resolve to 851 reviewed work groups, and the 14-pair duplicate queue is closed (§2.5), but many groups still need typed author, date, container, edition/locus, and access metadata before publication.
8. **Register scope decisions propagate.** The event-unit rule, inclusion scope, and audit judgments define what could be measured at all; a different rule would draw a different space. Full-roster application, source verification, score reconciliation, and alternative-resolution models were completed for the frozen releases, but they test the published rule rather than proving it uniquely correct. Those results do not validate post-freeze display overrides, unapplied corrections, or unresolved held events.
9. **Event-unit fairness is evidenced, not guaranteed.** Every active candidate row and relevant superseded parent has a recorded disposition, evidence trail, overlap treatment, and scoring path. No single row determines the ninth candidate direction, but program substitutions, date proxies, weighting, imputation, documentation, and source resolution remain material. These results support procedural auditability and bounded robustness; they do not prove that boundary bias has been eliminated or that every qualifying event has been found. The final Editorial Suite matched all 43 changed post-v5 claim groups and authorized this bounded wording for v6, subject to immutable release construction and validation.

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10. Conclusion

Documented land-taking and forced removal recur across widely separated times and places, in combinations whose similarities and differences warrant comparison. That is the historical claim of this selected-positive-case corpus, not a law of human nature. Without a representative denominator of encounters or opportunities for taking, it cannot establish universal human propensity, inevitability, or the rarity of peaceful alternatives.

The project set out to replace a hand-made taxonomy with a discovered one and ended somewhere more instructive: a persistent, replicable trait structure whose exact dimensionality is boundary-sensitive, and no tested automatic partition sturdy enough to hand categories to the public. In the historical 407→583 comparisons, a multidimensional trait space persisted across a 43% expansion; later v6 weighting and completion alternatives materially move the space (§5.2). The factor count crosses a documented eight/nine boundary as corpus composition and source resolution change; and every tested automatic attempt to make the events fall into self-evident kinds fails the project's stability and archive-density diagnostics. The defensible architecture is a division of labor — measurement proposes, review disposes: reviewed families for navigation, evidence-aware trait profiles for description, immutable releases for accountability, and a correction channel for everything the instrument gets wrong.

For comparative historical vocabulary, the specific findings are usable with an important limit: severity emerged as a largely distinct direction from the legacy mechanism variables in the factor solutions, with nonzero factor correlations, but the normalized mechanism instrument failed its blinded reliability gate. This is not an empirical vindication of a second public mechanism taxonomy. Within the measured trait space, "expulsion" and "internal colonization" name intents, not structures, and fragment when measured; whether anyone arrives to replace the removed is one of the strongest boundaries in the space; and the prehistoric expansions long treated as analogues of the real thing are among its most coherent instances. For method, the negative results are the

portable ones: in this corpus, automatic clusters weighted the past by its archives, and factor counts moved with documentation density — so a project that publishes cluster labels or "the k factors of history" without resolution-sensitivity and boundary diagnostics is publishing its source density with extra steps.

For practice, finally, the project is a working demonstration with stated uncertainties: a single scholar directing AI research agents under agreement-tested protocols, explicit decision ledgers, and immutable releases can build, measure, audit, and publish a worldwide register — provided the agents' work is gated, the failures are kept, the record survives outside the conversation, and every claim remains attached to a document a reader can check. The fact-check that produced this revision is part of that record.

AI contribution and disclosure statement

Instrument design (five independent disciplinary batteries), per-event source research, sixty-dimension coding, statistical analysis, sensitivity design, document drafting (including this monograph draft and its predecessor), and atlas build tooling were performed by large-language-model research agents from the Anthropic Claude family and OpenAI Codex (GPT-5–based), operating under written protocols with schema-enforced outputs. Sonnet 5 was the candidate coder and was **excluded** after failing the §4.2 gate; Fable 5 supplied the banked reference set; and Opus 5 produced the full coding matrix. Codex performed substantial second-pass source research, corpus and atlas integration, implementation, and independent audit and verification work. The independent fact-check of draft v0 and the later completion audit were likewise model-performed and are preserved alongside this draft. The documented consequential scope, inclusion, retirement, family-adoption, and publication decisions were reserved for or confirmed by the human author (per the project's decision ledgers; §4.5), who also set the constraints in §1. Model adjudicators resolved many coding disagreements under written no-averaging rules; final human responsibility attaches to the consequential editorial and publication decisions, not a claim of personal review of every cell. Agreement statistics and their limits are reported in §4. (Placement decided 2026-08-21: sole human author with this statement in back matter; venues that require the disclosure inside Methods can absorb it into §4 without change of content.)

Data and software availability

The register, codebook, coded matrices, missingness matrices, analysis scripts, sensitivity diagnostics, release manifests with SHA-256 hashes, integration ledgers, and this monograph's census generator and inventory are in the version-controlled project directory under `audit/`, `analysis/`, and `analysis/second_pass/releases/second_pass_v6_675/`. Git history and immutable hash-manifested releases provide local integrity; an archival deposit remains required before formal circulation. The publication destinations are the working monograph and checksum-bound v6 data. The current local atlas inventory is 370 cases and 380 playback stops, with a public feedback endpoint; the opening snapshot distinguishes its local build from live deployment. Frozen v6 downloads do not incorporate later display overrides or unresolved corrections merely because the atlas presents them.

References and background anchors (inventory in Appendix A)

Citation-resolution pass completed 2026-08-21: all original 40 entries verified against publishers or authoritative indexes — 36 correct without change, 4 corrected (Sokoloff & Engerman author order and title; Lemkin subtitle and publisher; Maier imprint; Scott subtitle), 0 unverifiable. Verdicts, checked URLs, and edition/variant cautions (Belich subtitle variant; Tilly 1990-vs-1992 editions): `monograph/CITATION_RESOLUTION.md`. Malhi et al. 2008 was added on 2026-08-27 from the opened full-text PMC record and DOI metadata used in the event-unit verification. Thirteen original entries remain background anchors rather than in-text citations; a final circulation bibliography should either cite them substantively or move them to a separately labeled background list.

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Appendix A. Source inventory

`monograph/bibliography_census.csv` — 854 unique URL records with provenance tags, regenerated by the saved `monograph/build_bibliography_census.py` (URL-normalization dedup rule documented in the script; 845 at the fact-check checkpoint). The reviewed work-level output is `monograph/bibliography_works.csv`: 851 groups after all 14 near-title relationships were adjudicated. Method and dispositions: `monograph/BIBLIOGRAPHY_REVIEW.md` and `monograph/bibliography_review_queue.csv`.

Appendix B. (planned) Codebook

The sixty dimensions with anchors, evidence notes, and era-coverage rules — currently `analysis/codebook/codebook.md` verbatim.

Appendix C. (planned) Agreement, stability, and sensitivity tables

Pilot agreement; model-gate comparison; cross-model agreement including N/A concordance; factor-count boundary diagnostics; loading congruence tables; cluster stability and archive-density sensitivity; era-reweighting cosines; event-boundary reviewer agreement; parent/child overlap coverage; and equal-event versus grouped-program, parent/successor, era-, region-, mechanism-, duration-, and documentation-resolution sensitivity. All currently in `analysis/`, `analysis/validation/`, and `analysis/second_pass/results/`.